Name_____ Date_____ Period_____

LESSON 2-2 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Building a Histogram

Maryland Standard 6.SP.B.4 · Learning target – I can group data into equal intervals and build a histogram from a frequency table.

REMEMBER

- A **histogram** groups data into **equal intervals**, like 0–9, 10–19, 20–29.
- The **frequency** is how many values fall in an interval. It sets the bar height.
- Intervals do not overlap, and the bars **touch**.

WORKED EXAMPLE

Minutes of practice: 4, 15, 22, 9, 18, 27, 13, 16, 21, 11. Make a frequency table.

Step 1 0–9: 4, 9, so the frequency is 2.

Step 2 10–19: 15, 18, 13, 16, 11, so it is 5.

Step 3 20–29: 22, 27, 21, so it is 3.

Answer **2, 5, 3**

WORD BANK

interval – a group of numbers, like 10–19

frequency – how many numbers are in a group

histogram – a graph with bars that touch

overlap – share the same number

PRACTICE

show your work · label units

1. **Points Scored.** A player scored these points in 12 games: 8, 14, 21, 5, 17, 12, 25, 19, 11, 23, 16, 9. Complete the frequency table.

Step 1 Find the numbers from 0 to 9. Count them.

Step 2 Do the same for 10–19 and 20–29.

Step 3 Check: the counts add up to 12.

Points	Tally	Frequency
0–9		
10–19		
20–29		

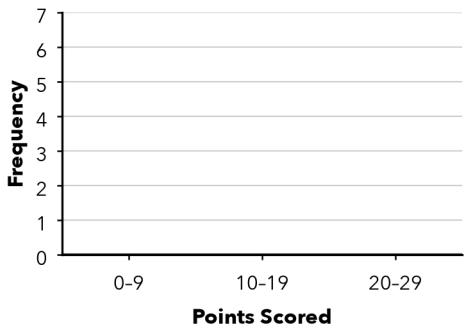
2. Draw the histogram for Problem 1. Bars should touch.

Step 1 Use the counts from Problem 1 for the bar heights.

Step 2 Draw the bars so they touch.

a. Which interval has the tallest bar? _____

b. Games with 10 or more points: _____



3. **Picnic Ages.** 34, 22, 41, 38, 27, 45, 31, 36, 29, 48 years. Find each frequency.

a. 20–29: ____ **b.** 30–39: ____ **c.** 40–49: ____

4. **Missing Bar.** A histogram shows **24** students. Three bars are **5**, **9** and **3** tall. There is one more bar. How tall is it?

Step 1 Add the three bars you know.

Step 2 Subtract that sum from 24.

Answer_____ students

5. **Find the Error.** Marcus used the intervals 0–9, 10–14, and 15–29. Explain his mistake and fix it.

Frame *Marcus's intervals are not ___. Better intervals are ___, ___ and ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Ask family members how many minutes they read today. Group the answers in intervals of 10.

Name_____ Date_____ Period_____

LESSON 2-2 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Describing Histogram Shape

Maryland Standard 6.SP.B.4 · Learning target – I can describe the shape of a histogram using its peak, spread, and balance.

REMEMBER

- **Peak**: the tallest bar. **Symmetric**: sides match.
- **Skewed right**: the tail stretches right. **Skewed left**: the tail stretches left.
- A histogram shows counts, not exact values.

WORKED EXAMPLE

Bars of height 2, 7, 4, 2, 1 over 0–9, 10–19, 20–29, 30–39, 40–49. Shape?

Step 1 The peak is 10–19, near the left.

Step 2 The bars shrink to the right: a tail on the right.

Answer **Skewed right**

WORD BANK

peak – the tallest bar

symmetric – both sides match

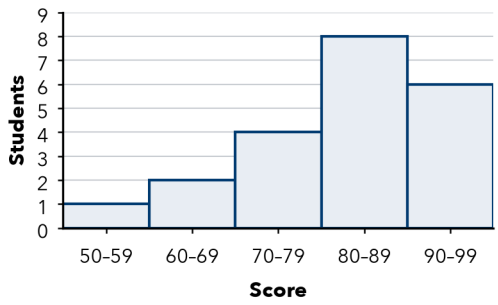
skewed right – long tail on the right

skewed left – long tail on the left

PRACTICE

show your work · label units

1. Test Scores. The histogram shows the scores on a class test.



Step 1 Add all the bar heights for a.

Step 2 The tallest bar is the peak.

Step 3 Look for the long tail to name the shape.

a. Number of students: _____

b. Peak interval: _____

c. Shape: _____

d. Scores of 80 or more: _____

2. Bar heights are listed from left to right. Name each shape.

Step 1 Find the biggest number in each list.

Step 2 Look at which side has the small numbers: that is the tail.

a. 1, 4, 8, 4, 1 _____ **b.** 9, 6, 3, 2, 1 _____ **c.** 1, 2, 3, 6, 10 _____

d. 3, 5, 9, 5, 3 _____ **e.** 12, 8, 5, 3, 2 _____ **f.** 2, 3, 5, 8, 12 _____

3. **Race Times.** **30** runners. Bars of **3, 8, 11, 6, 2** show times of 20–24, 25–29, 30–34, 35–39, 40–44 minutes.

   a. How many runners finished in less than 30 minutes? _____

   b. How many finished in 35 minutes or more? _____

4. **Stretch.** Write 5 bar heights, from left to right, for a histogram of **20** values that is **skewed left**.

   Bar heights_____

5. **Explain.** Ramon says the Test Scores histogram shows a score of 85. Is he right? Explain.

   Frame *Ramon is ___ because a histogram only shows ___.*

TEST PRACTICE

choose the best answer

6. Bar heights are listed from left to right. Which histogram is skewed right?

   A 2, 5, 9, 5, 2 **B** 10, 7, 4, 2, 1 **C** 1, 2, 4, 7, 10 **D** 5, 5, 5, 5, 5

7. Bars of height 4, 7, and 5 sit over 0–9, 10–19, and 20–29. How many values are 10 or more?

   A 7 **B** 11 **C** 12 **D** 16

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___